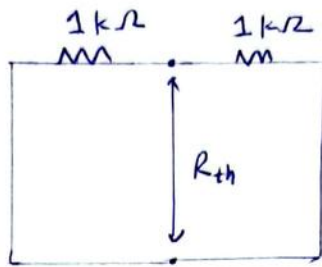
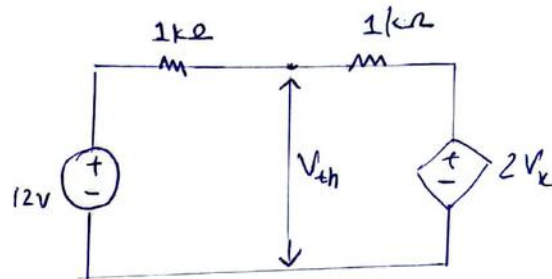


Q5) Using thevenine theorem $2k\Omega$ is load resistance for R_{th}



$$R_{th} = \frac{1}{\frac{1}{1} + \frac{1}{1}} = \frac{1}{2} k\Omega = 0.5 k\Omega$$

for V_{th}

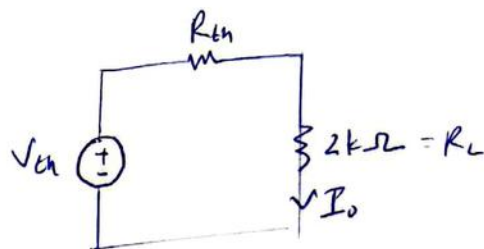


current in loop : loop equation

$$12 - I - I - 2I = 0 \Rightarrow 12 - 3I - I = 0 \Rightarrow I = 3 \text{ mA}$$

$$V_{th} = 12 - 1(3) = 9 \text{ V}$$

so circuit is



$$I_0 = \frac{V_{th}}{R_{th} + R_L} = \frac{9}{0.5 + 2} = 3.6 \text{ mA}$$

$$\boxed{I_0 = 3.6 \text{ mA}}$$

Q3)

3.1: H_0 : There is no difference in average number of hours spent on homework between girls and boys

H_1 : high school girls spend more hours per week on their homework

3.2: • The populations from which samples are drawn are normally distributed

• The variances of the two populations are equal

3.3: less, reject, have

Q1)

1.1: Sample proportion = $\frac{X}{n} = \frac{15}{790} = 0.01899$

1.2 40% confidence interval

$$\alpha = 1 - 0.4 = 0.1$$

critical value: $Z_{\frac{\alpha}{2}} = Z_{0.05} = 1.645$

confidence interval: $\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$$: 0.01899 \pm 1.645 \sqrt{\frac{0.01899(1-0.01899)}{790}}$$

$$= (0.0110, 0.02698)$$

1.3: $n = \frac{Z^2 p(1-p)}{E^2} = \frac{1.645^2 \times 0.01899 \times (1-0.01899)}{0.03^2}$

$$n = 56$$

Q6) FBD of block B

$$-P \cos 30^\circ + \frac{5}{13} N_B + \frac{12}{13} \times 0.5 N_B = 0 \Rightarrow N_B = \frac{P \cos 30^\circ}{\frac{5}{13} + \frac{12}{13} \times 0.5}$$

$$= 1.023 P$$

$$-550 - P \sin 30^\circ + N_B \times \frac{12}{13} - \frac{5}{13} \times N_B \times 0.5 = 0$$

$$0.2476 P = 550 \Rightarrow P = \frac{550}{0.2475} = 2222.2 \text{ lb}$$

$$\boxed{\text{Part B: } P = 2.22 \text{ kip}}$$

$$\text{Normal force at B } N_B = 1.023 \times 2222.22 = 2273.33 \text{ lb}$$

FBD of block A

$$-\frac{12}{13} \times N_B + \frac{5}{13} \times 0.5 \times N_B - 550 + N_A = 0 \Rightarrow N_A = 2211.3 \text{ lb}$$

$$-\frac{5}{13} N_B - \frac{12}{13} \times 0.5 N_B + 0.2 \times N_A + k_s = 0 \Rightarrow s = 1.85 \text{ ft}$$

$$\boxed{\text{Part A: } s = 1.85 \text{ ft}}$$

Q2]

2.1: $H_0: \mu = 4$, $H_1: \mu > 4$

2.2: The assumption is that the distribution of battery life follow a normal distribution

2.3 Test statistic $= \frac{4.05 - 4}{0.2/\sqrt{50}} = 1.7678$

2.4 critical value: from tables for one tailed test critical value is 1.676

2.5: more, reject, have

$$\begin{aligned}
 I_{y1} &= \frac{2b \times (2a)^3}{12} + A_1 (0-x_1)^2 \\
 &= \frac{2 \times 150 \times 8 \times 140^3}{12} + 84000 (0-140)^2 \\
 &= 2.1952 \times 10^9 \text{ mm}^4
 \end{aligned}$$

$$\begin{aligned}
 I_{y2} &= \frac{\pi}{64} D^4 + A_2 (0-x_2)^2 \\
 &= \frac{\pi}{64} (150)^4 + 17671.458 (0-140)^2 \\
 &= 0.371211 \times 10^9 \text{ mm}^4
 \end{aligned}$$

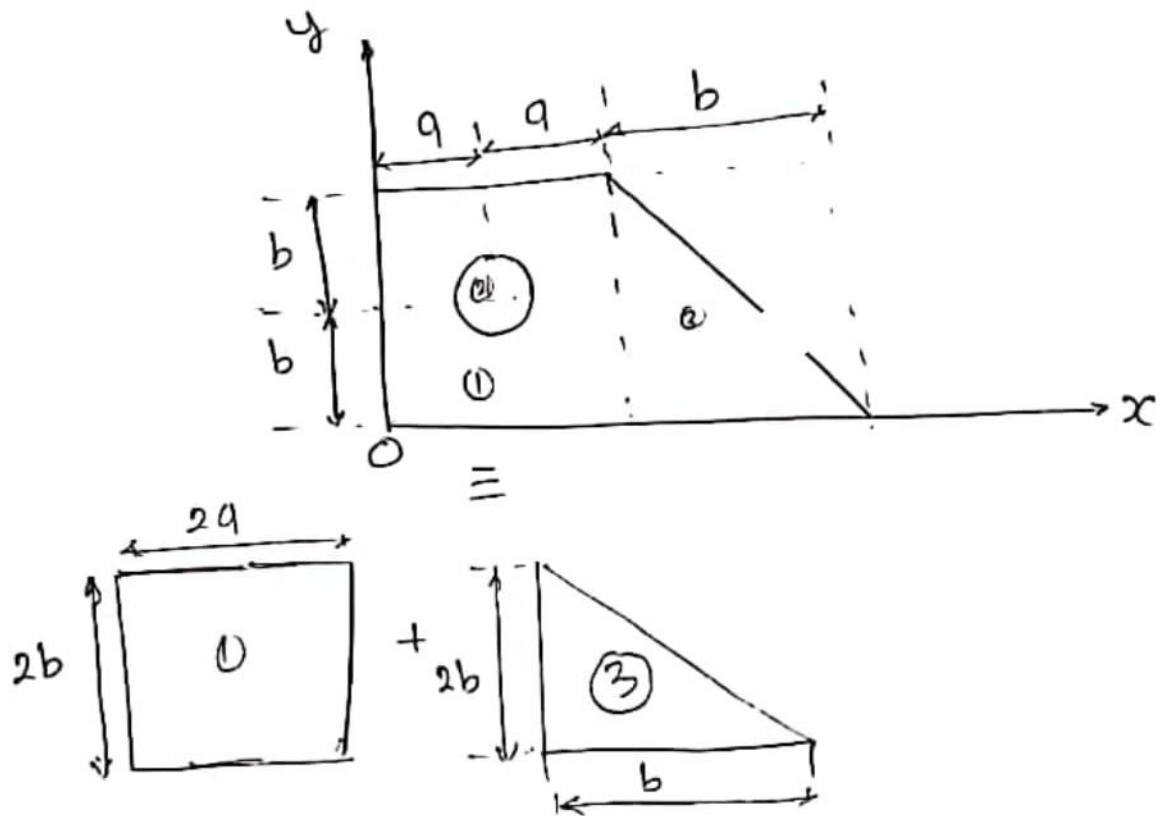
$$\begin{aligned}
 I_{y3} &= \frac{b^3 \times 2b}{36} + A_3 (0-x_3)^2 \\
 &= \frac{150^3 \times 2 \times 150}{36} + 22500 (0-330)^2 \\
 &= 2.478375 \times 10^9 \text{ mm}^4
 \end{aligned}$$

$$\begin{aligned}
 I_y &= I_{y1} - I_{y2} + I_{y3} \\
 &= 2.1952 \times 10^9 - 0.371211 \times 10^9 + 2.478375 \times 10^9
 \end{aligned}$$

$$I_y = 4.30236 \times 10^9 \text{ mm}^4$$

$$\boxed{\text{Part A: } 4.30 \times 10^9 \text{ mm}^4}$$

Q5)



$$A_1 = 2a \times 2b = 4ab = 4 \times 140 \times 150 = 84000 \text{ mm}^2$$

$$x_1 = \frac{2a}{2} = a = 140 \text{ mm}$$

$$A_3 = \frac{1}{2} \times b \times 2b = b^2 = 150^2 = 22500 \text{ mm}^2$$

$$x_3 = 2a + \frac{b}{3} = 2 \times 140 + \frac{150}{3} = 330 \text{ mm}$$

$$A_2 = \pi r^2 = \pi \times 75^2 = 17671.458 \text{ mm}^2$$